Umar Ghani

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Education

University of California, Berkeley — Berkeley, CA
Bachelor of Arts in Data Science; Minors in Computer Science and Economics — *Expected May 2028*

Experience

Berkeley Artificial Intelligence Research (BAIR) Lab — August 2026 – Present
Researcher — *Berkeley, CA*

- Conducting research on LLM evaluations, advised by PhD researcher Mert Cemri under Prof. Ian Stoica.
- Exploring adaptive evaluation and improvement methods for multi-agent systems.

Qualcomm — Summer 2026
Software Engineering Intern — *San Diego, CA*

- Developed an AI-powered test generation platform to optimize 5G NR and LTE modem validation.
- Reduced engineering test case creation time by 50% through automated drafting and workflow automation.
- Built and deployed a scalable full-stack web application using React, Django, Docker, and Kubernetes.
- Designed robust OpenAPI-compliant FastAPI services to automate protocol and format verification.
- Migrated legacy ASN.1 test cases and low-level protocol definitions to a modern test framework.

NASA Ames Research Center — January 2026 – May 2026
Researcher — *Moffett Field, CA*

- Researched and built Vision Transformer and ConvLSTM models for early solar active region detection.
- Utilized multi-modal satellite imagery including magnetograms and Dopplergrams for model inputs.
- Applied the Hungarian algorithm and seed region growth to track complex solar regions over time.
- Trained deep learning models to predict active region evolution on NASA high-performance computing resources.

Projects

When Agents Move Money: Jailbreaking Financial-Institution AI Agents | *Python, DTAP, Docker, RAG*

- Engineered an automated blue-teaming agent on Virtue AI's DTAP to patch agent harness vulnerabilities.
- Analyzed red-team attack traces to generate prompt guardrail patches and RAG security playbooks.
- Sandboxed isolated agent environments to safely simulate and learn from complex indirect prompt injections.
- Evaluated zero-shot defenses on financial domain models, reducing Attack Success Rates while preserving utility.

DrystAI – Finalist @ Google Gemini Hackathon | *MongoDB Atlas, Google Gemini, WebRTC, Next.js*

- Built an AR networking assistant that recognizes individuals and recalls past interactions in real time.
- Utilized Gemini Vision, WebRTC, and Speech APIs to enable live multimodal interaction tracking.
- Placed as a Top 6 Finalist out of 59 teams at the Google Gemini Hackathon held at Shack15.

Nimverse Security – 1st Place @ NVIDIA Hackathon | *NVIDIA DGX Spark, PyTorch, React, Next.js*

- Developed Nimverse Security Monitor for real-time multimodal emergency detection from data streams.
- Trained deep learning models using PyTorch on NVIDIA DGX Spark with SF city camera feeds.
- Optimized processing pipelines for low-latency inference and real-time streaming deployment.
- Won 1st Place in the Nemotron challenge track at the NVIDIA Spark Hackathon.

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, C#, C, HTML/CSS
Frameworks: React, Next.js, Node.js, Django, FastAPI, Flask
AI / ML: PyTorch, Transformers, LLMs, LLM Evaluation, Multi-Agent Systems, RAG, Computer Vision
Databases / Data: PostgreSQL, Redis, MySQL, NumPy, pandas, Scikit-learn
Systems / Infrastructure: Docker, Kubernetes, GCP, REST APIs, OpenAPI, WebRTC, Git, System Design